

HELIX ANALYZER

How to Use — DNA Sequence Analysis Guide

Quick purpose: Helix Analyzer lets you paste or upload a DNA sequence and inspect its composition, translation, ORFs, restriction sites, motifs, and other sequence properties directly in the browser.

Workflow	Paste / Upload DNA → Validate → Analyze → Explore 7 tabs → Copy / Export
Input	Raw DNA, FASTA, sample sequences, or demo sequence
Output	Sequence statistics, protein translation, ORFs, restriction sites, motifs, and tools

1. Getting Started — Load a Sequence

- Paste manually: Click the main text box and paste raw DNA or FASTA text.
- Upload FASTA: Click  Upload FASTA and select a .fasta, .fa, .fna, or .txt file.
- Load a sample: Use β -globin CDS, Plasmid MCS, or Random 600 bp.
- Load Demo Sequence: On the landing page, click the hero button to auto-fill and analyze a demo sequence.

Raw DNA example

```
ATGGTGCATCTGACTCCTGAGGAGAAGTCT...
```

FASTA example

```
>my_gene description here
ATGGTGCATCTGACTCCTGAGGAGAAGTCT...
```

The tool automatically ignores line breaks, spaces, and numbers; converts U → T; detects invalid characters; and reads the FASTA header as the sequence name.

2. Verify Input — Live Feedback

As you type, the input status updates immediately.

Status	Meaning
 Awaiting input	No sequence has been entered.
 N invalid character type(s)	The sequence contains unsupported characters.
 Too short	Fewer than 3 bases are present.
 Valid DNA · GC 52.3%	The sequence is ready for analysis.

The base-count badge also shows the total number of bases, for example: 1,247 bases.

3. Run the Analysis

Click  Analyze Sequence to process the current sequence. You can also press Ctrl + Enter on Windows or Cmd + Enter on Mac while inside the text box.

After analysis, the results section displays six summary stat cards and seven analysis tabs.

4. Understand the Results

Tab	What it does
Overview	Shows the GC Content Profile, a color-coded Sequence Viewer, and sequence details such as purine
Composition	Shows A/T/G/C counts and percentages, plus a codon-usage table sorted by frequency.
Translation	Converts DNA to an mRNA transcript (T → U), displays the frame +1 protein sequence, and provides
ORF Finder	Scans all six reading frames for possible open reading frames. Set the minimum amino-acid length
Restriction Map	Displays restriction-enzyme cut sites visually and in a table with enzyme, recognition sequence, cou
Motif Search	Searches for exact motifs or IUPAC patterns. Search both strands to find hits on the forward and rev
Tools	Provides reverse complement, complement, reversed sequence, and GC skew per 100-bp window.

5. Stat Cards

The top summary row provides:

- Length: Total number of bases.
- GC%: Percentage of G and C bases.
- AT%: Percentage of A and T bases.
- Molecular Weight: Approximate molecular weight of the DNA.
- Melting Temperature: Estimated DNA melting temperature.
- Ambiguous Bases: Count of uncertain bases such as N.

6. ORF Finder — Practical Use

An ORF (Open Reading Frame) is a candidate protein-coding region. Enter the minimum amino-acid length, then click Find ORFs. The analyzer scans six frames: +1, +2, +3 and -1, -2, -3.

The results table reports strand, frame, start/end position, length, and a protein preview.

7. Restriction Map

Restriction enzymes recognize specific DNA sequences. For example, EcoRI recognizes GAATTC. Use the Restriction Map to see whether an enzyme has sites in your sequence and where those sites occur.

The map provides a visual cut-site bar plus a table containing enzyme name, recognition site, number of sites, and positions.

8. Motif Search & IUPAC Codes

Enter an exact motif such as GAATTC, or an IUPAC pattern such as TATAWAW. Then choose Search Both Strands to search both orientations.

Code	Meaning	Code	Meaning
A	A	R	A or G
T	T	Y	C or T

Code	Meaning	Code	Meaning
G	G	S	G or C
C	C	W	A or T
N	Any base	K	G or T
M	A or C	B	C/G/T
D	A/G/T	H	A/C/T
V	A/C/G		

Example: TATAWAW allows W to match A or T, so it can find TATA-box-like patterns.

9. Copy & Export

Action	How
Copy a result block	Click the Copy button for that block.
Copy the whole sequence	Open Overview → Copy.
Download report	Click ↓ Report in the top-right to save a .txt report.

10. Quick Workflow Example

1. Click Load Demo Sequence on the hero page.
2. Scroll to the results and review length, GC%, and molecular weight.
3. Open Translation and copy the protein sequence.
4. Open ORF Finder, set minimum length to 50 amino acids, and run the scan.
5. Open Restriction Map and check for EcoRI sites.
6. Open Motif Search and enter AATAAA to look for that motif.
7. Click ↓ Report to save the analysis report.

11. Keyboard Shortcuts

Shortcut	Action
Ctrl / Cmd + Enter	Run analysis
Enter in ORF minimum box	Re-run ORF scan
Enter in motif box	Run motif search

12. One-Line Summary

Paste or upload DNA → click Analyze → browse the 7 tabs → copy or download what you need.

All analysis is designed to run in the browser, so according to the tool description, sequence data does not leave the device.

Note: Outputs such as ORFs, motifs, melting temperature, and restriction sites are computational results and should be interpreted in the appropriate biological context and verified with suitable reference tools before

experimental decisions.